

Chapter 11

A Quantum Journey

Exploring the theory, building the future

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Quantum computing was not about a fascination with physics or an abstract curiosity about the behaviour of particles. As a self-confessed ‘geek’ with an interest in computing and later cybersecurity. My interest in quantum derived as seeing it as a security problem. Reading articles on Post Quantum Cryptography (PQC) I realised we had a technology that could break some existing cryptography and with that technology. As I dived deeper in my PhD in Computer Science I realised that quantum computers could also be attacked with an assortment of Denial of Service (DoS) attacks this then started me on my journey with Q-SLICE and QUANT

My current role as a security architect has seen me working on a variety of systems assessing and adding security which included threat modelling, risk governance and system benchmarking. Adding compensating controls to end of life or OT systems. This has helped me have a wider and broader view of systems and balancing useability with security and business requirements as they don’t always meet in the middle.

Quantum computing presented itself precisely in that way. It wasn’t a technology I approached with admiration or excitement. It was new technology which I approached with a security lens, forming slowly under the cryptographic foundations I have always had an interest in. The more I researched quantum the clearer it became that quantum wasn’t simply another emerging technology to monitor it was a fundamental shift in computational capability with direct implications for confidentiality, integrity and long term trust.

I have always wanted to understand the ‘how’ something works and ‘why’ it works that way. This is the same approach I took with quantum computing. Once I understood how it worked, I needed to understand how it changed adversarial capability. How it

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reshaped risk horizons and how it undermined current security. By understanding the adversarial element, I could then start to think about the defensive side and the protection that would be required. After all security is a protective and defensive discipline.

My early career didn't begin in industry; it began in the classroom. I started my career as a computer science teacher in both school and college environments, helping students understand the fundamentals of computing, logic and system design. Teaching programming, web development, networking and system support at various levels and age groups. Those years shaped the way I approach technology. Not as a collection of tools, but as a set of concepts that must be understood, explained. Understanding the foundational concepts to build on the more complex elements. These same foundational concepts are what underpin security especially in the more technical world of cybersecurity. Teaching helped me to break complexity into clarity and that skill would become essential later in my journey. By teaching something you also learn it in detail and deepen your own understanding of the topic or subject. You also get interesting questions, different points of view and understanding. I have always found I learn just as much teaching as my learners do!

From education, I moved into policing as a police IT instructor where I had been volunteering for many years as a police officer. That shift brought me into not only operational IT and systems training but into the world of Open Source Investigation (OSI) and the development of a cybercrime course for detectives. Working with SME's in digital forensics and open source I learnt an awful lot about cybercrime, which led me to the world of cybersecurity. However, it was cybercrime, not cybersecurity, that first taught me the mindset of the adversary. It was the opposite side of the coin, but it completed the picture when I moved into teaching cybersecurity.

This experience led me deeper into the security profession. I transitioned into instructing for major security certification bodies such as CompTIA, EC-Council, ISACA and ISC2. Where I taught practitioners how to defend systems, govern risk and build resilient architectures. Each certification I delivered I had to pass the certification before being able to deliver it. This enables me to soak about a wealth of information across twenty certifications. My focus during this period was on classical security such as cryptographic hygiene, governance frameworks, adversarial modelling, secure networks, logging and monitoring and the operational realities of protecting data. Moving into industry enabled me to take the theory and apply it, with a realisation that sometimes the theory does not work in practice and is either not useable or has to be adapted and changed. It was during my time at Vodafone working on a control roadmap that I first came across PQC, which led to me into the world of quantum and quantum security.

Quantum computing reframed the entire problem. Even before practical quantum machines existed, the *harvest-now, decrypt-later* (HNDL) threat model made it clear that adversaries didn't need a quantum computer today to break tomorrow's secrets. They only needed storage and patience. The threat wasn't hypothetical it was already operational. Data with long term value was already being collected, archived and stockpiled by actors who understood exactly what was coming.

This was the pivot point for me. Quantum wasn't a future technology to monitor

or a distant research topic to keep an eye on. It was a present day risk, exerting pressure on cryptographic assumptions that had held for decades. My background from teaching fundamentals, to understanding adversarial behaviour in policing, to training professionals in cybersecurity and working on large security projects. Meant I recognised the pattern immediately. Quantum was a structural shift in the threat landscape, and it demanded a new way of thinking and adapting or changing current security thinking and concepts for this new opportunity and threat.

11.1 Learning Quantum: A Pragmatic, Security First Approach

My entry into quantum computing didn't begin with wavefunctions, Hilbert spaces or the mathematical elegance that physicists might admire. I didn't come to quantum through the door of theoretical physics at all never mind via computer science. I approached it the same way I approached every emerging technology. As a teacher, security architect and adversarial thinker looking for how it worked and then how quantum could be used to attack or be attacked.

My inquisitive nature and wanting to understand the depth as well as the breadth led me to ask myself:

- What can it compute that classical systems cannot?
- How does it impact on cryptographic systems?
- What does quantum bring that we have not got already?
- How could it be used by adversaries?

These questions shaped my learning journey. More so with the initial research I was undertaking with my PhD looking at quantum computing from a security standpoint and what quantum security should look like. To do this I needed to understand quantum computing as a computational model with different scaling behaviours, one that alters the economics of attack and defence.

Once quantum computation is treated as controlled interference rather than abstraction, its impact is explicit. Shor's algorithm directly compromises RSA, ECC and all public-key systems built using this type of cryptography. Grover's algorithm also applies a pressure to symmetric cryptography, compressing effective key strength and invalidating legacy comfort around 128-bit targets.

The more I learned, the more obvious it became that quantum computing wasn't a scientific curiosity. It was also a threat vector with a new class of adversarial capability that demanded the same scrutiny, modelling and governance as any other high impact security risk. My learning path was therefore pragmatic, not theoretical. I focused on attack surfaces, cryptographic fragility and the real world implications of quantum advantage. However narrow or specialised it may be. Although without the same level of real world data due to quantum still being largely around theory as opposed to practical application and knowledge.

Quantum computing is best understood as the manipulation of probability amplitudes rather than classical bits. Classical computers operate on definite states, a bit is either 0 or 1. Quantum systems, by contrast, evolve over superpositions. Weighted combinations of 0 and 1, where each component carries a complex probability amplitude. These amplitudes can interfere with one another, reinforcing some outcomes while cancelling others. This interference behaviour is the source of quantum computational advantage. Despite this richer internal structure, any measurement of a qubit still yields a definite classical result: 0 or 1. The strength of quantum computing doesn't come from science-fiction notions of infinite parallelism. It comes from three very specific physical phenomena that translate into computational advantages:

- **Superposition:** the ability to represent many possible states compactly within a single quantum register. This isn't "doing everything at once"; it's encoding a landscape of possibilities in a way classical systems cannot.
- **Entanglement:** the ability to correlate qubits so strongly that the state of one instantly constrains the state of another, no matter how far apart they are. In computational terms, entanglement creates relationships between data elements that classical systems cannot replicate. Qubits do not share a state; they share a constraint on possible joint outcomes.
- **Interference:** the mechanism that allows quantum algorithms to amplify the probability of correct solutions while suppressing incorrect ones. This is the real engine of quantum advantage: shaping the probability landscape so that the right answers become more likely.

These principles are elegant, but their relevance to security is brutally practical. Quantum computers do not threaten cryptography because they are "faster." They threaten cryptography because they compute differently. They exploit mathematical structures in ways classical machines fundamentally cannot.

Shor's algorithm doesn't succeed because it runs quickly; it succeeds because it uses interference to expose periodicity in number-theoretic problems that underpin RSA and ECC. Grover's algorithm doesn't magically accelerate brute force; it reshapes the search landscape so that the correct key becomes quadratically easier to find.

Once you understand quantum computing as a fundamentally different computational model, with its own scaling behaviours and operational principles. The security implications become impossible to ignore. Quantum systems are not faster versions of classical computers; they implement a distinct method of computation with unique strengths and weaknesses. They do not replace classical architectures but instead provide specialised speedups for particular problem classes that are intractable or slow on classical machines.

From a security perspective quantum isn't just an interesting computational model. It introduces new adversarial capabilities, breaks long standing assumptions and forces organisations to rethink how they protect data across its full lifecycle. The threat is not singular or isolated. It is multidimensional.

Quantum computing exerts pressure in three major areas:

Cryptographic fragility

The most immediate and widely discussed impact is the fragility of classical public-key cryptography. Algorithms that have underpinned secure communication for decades such as RSA, ECC and Diffie-Hellman, These are mathematically vulnerable to Shor’s algorithm. This isn’t a theoretical weakness; it is a structural one. Once quantum machines reach sufficient scale and stability, these algorithms collapse.

Symmetric cryptography fares better as Grover’s algorithm effectively halves the security strength of symmetric keys, forcing organisations to double key sizes to maintain equivalent protection. AES-128 becomes insufficient. AES-256 becomes the new baseline.

Temporal asymmetry

Quantum also introduces a profound shift in how we think about confidentiality over time. In classical security models, encryption protects data at the moment it is transmitted or stored. Once encrypted, it is assumed to remain confidential for as long as the cryptographic scheme remains secure.

11.2 Quantum Breaks This Assumption

Adversaries can and already do harvest encrypted data today with the intention of decrypting it in the future once quantum capabilities mature. This “harvest-now, decrypt-later” strategy turns long-lived data into a liability. Sensitive information with multi decade value medical records, genomic data, state secrets, intellectual property, financial archives become vulnerable long before quantum computers exist at scale.

Harvest Now, Decrypt Later (HNDL)

HNDL describes the collection of encrypted data today with the intention of decrypting it once large-scale quantum computers can break classical public key encryption. This is a confidentiality centric threat: the attacker’s objective is to expose information that is currently protected by RSA, ECC, or other vulnerable schemes. HNDL matters most for:

- Long-retention data
- Sensitive archives
- Communications requiring multi-decade secrecy
- National-security and diplomatic material

HNDL is widely recognised, but it is only the beginning.

Harvest Now, Forge Later (HNFL)

HNFL shifts the target from encryption to digital signatures. Instead of collecting ciphertext, adversaries collect signed artefacts software updates, firmware images, certificates, audit logs, legal documents, financial transactions with the intention of forging signatures once quantum computers can break classical signature schemes. HNFL is an integrity threat. It enables:

- Forged software updates
- Counterfeit certificates
- Manipulated audit trails
- Fraudulent transactions
- Retrospective rewriting of “trusted” records

Where HNFL compromises secrecy, HNFL compromises the truth of digital artefacts.

Trust Now, Forge Later (TNFL)

TNFL extends HNFL into the domain of trust infrastructure. Systems today rely on signatures, certificates, and identity proofs that will become forgeable in the future. TNFL captures the risk that trust anchors valid today become counterfeit tomorrow. TNFL is an authenticity and trust-chain collapse threat. It enables:

- Impersonation of devices, services, or authorities
- Fabrication of entire PKI chains
- Subversion of firmware-update ecosystems
- Undetectable supply-chain compromise
- Manipulation of safety-critical systems that rely on signed commands

TNFL is particularly dangerous for operational technology (OT), where trust in signed commands directly affects physical processes.

Deploy Now, Exploit Later (DNEL)

DNEL addresses a different dimension, lifecycle exposure. Many devices deployed today IoT sensors, industrial controllers, medical implants, EV chargers, automotive ECUs will remain in service long after quantum computers can break their embedded cryptography. If these devices cannot be upgraded to post-quantum primitives, they become permanently vulnerable.

DNEL is a cyber-physical safety threat. It enables:

- Impersonation of devices that cannot rotate keys
- Compromise of long lifecycle infrastructure
- Persistent footholds in OT environments

- Attacks on vehicles, energy systems, and medical devices
- Exploitation of unpatchable hardware trust anchors

DNEL is the most structurally difficult threat to mitigate because it arises from design choices made years before quantum exploitation becomes possible.

Governance and migration risk

The transition to post-quantum cryptography (PQC) is not a software update. It is a systemic migration that touches every layer of the digital ecosystem. Every protocol, every embedded device, every certificate chain, every key exchange mechanism, every cryptographic library. All must be re-evaluated and in many cases, replaced. This introduces governance challenges on a scale the industry has not faced since the early days of the internet. Organisations must:

- inventory cryptographic assets
- identify long lived data
- assess protocol dependencies
- test PQC performance and interoperability
- plan phased migrations
- manage hybrid cryptographic environments
- ensure supply chain readiness

The complexity is immense. The risk of misconfiguration, partial migration or inconsistent implementation is high. And because cryptography is deeply embedded in systems, the migration will take years, not months. The identification of cryptographic assets alone is a huge task for some organisations.

Building Q-SLICE and benchmarking quantum reality

As my understanding of quantum computing deepened, it became clear that the field suffered from a chronic imbalance with an element of hype and far too little measurement. Claims about “quantum advantage,” “breakthroughs” and “revolutionary performance” were often made without reproducible evidence, consistent benchmarking or any meaningful connection to real-world security impact. Security cannot be built on marketing language or theoretical potential. It must be grounded empirical behaviour. With little to know data with quantum still in its early stages an element of assumptions has to be made. Which ultimately when dealing with uncertainty within security is what an organisation must undertake. However, these assumptions are often built around data and knowledge assimilated over years.

This realisation led me to develop the Q-SLICE a threat model modular designed to look at quantum hardware and quantum-inspired systems with the same rigour we apply to classical security architectures using threat models like STRIDE. Q-SLICE

was built to answer a simple but critical question. What does quantum hardware actually do under load, under noise and under cryptographic pressure?

To answer that, I created a Q-SLICE Threat Harness using QSKIT modules as a small program to a range of foundational characteristics both in a simulator and on IBM Quantum Cloud. This covered elements such as:

- **Coherence** how long qubits retain usable quantum states before decoherence destroys computational value.
- **Fidelity** the accuracy of operations, gates and circuits, and how closely real hardware behaviour matches theoretical expectations.
- **Noise behaviour** the patterns, sources, and propagation of noise across circuits, and how noise interacts with algorithmic structure.
- **Scaling characteristics** how performance, error rates and resource requirements evolve as circuits grow in depth and width.
- **Backend variability** the differences between quantum hardware providers, architectures, and runtime environments, and how these differences affect reliability.
- **Cryptographic workload fragility** how quantum systems behave when executing circuits derived from cryptographic primitives, including those relevant to Shor-style and Grover-style attacks.

11.3 The Purpose of All This is Straightforward

Replace hype with measurement. Replace assumptions with evidence. Quantum security cannot be governed by optimistic projections, vendor claims or theoretical limits that assume perfect qubits and noiseless circuits. It must be governed by what hardware actually does. Its fragility, its variability, its scaling behaviour and its real-world impact on cryptographic workloads. Q-SLICE exists to bring clarity to a field clouded by speculation. It provides the empirical grounding needed to make informed decisions about risk, migration and long-term security strategy.

The human side: learning, misconceptions, and the reality of quantum

As I progressed deeper into the quantum domain, I quickly realised that the technical challenges were only half the story. The other half, often the more difficult half, was navigating the misconceptions, oversimplifications and confident misunderstandings that surround quantum computing. The field attracts hype in a way few technologies do AI is another with similar issues. I encountered the same recurring beliefs everywhere: in classrooms, in industry discussions, in vendor presentations and even in policy conversations.

“Quantum computers will solve everything”

This idea collapses the moment you examine what quantum machines are actually good at. Quantum computing is powerful, but it is narrowly powerful. It excels at specific classes of problems. Optimisation, simulation and certain algebraic structures but it is not a universal accelerator. Most workloads will never benefit from quantum hardware and many will perform worse.

“Quantum is decades away”

This belief persists because people equate “breaking RSA” with “quantum maturity.” But quantum risk doesn’t begin at the moment RSA falls. It begins the moment adversaries start harvesting encrypted data for future decryption. It begins when organisations realise their long lived data will outlast their cryptography.

“PQC is just a drop in replacement”

This is perhaps the most dangerous misconception. Post quantum cryptography is not a simple swap of algorithms. It changes key sizes, message sizes, handshake patterns, performance characteristics and protocol assumptions. It affects embedded systems, constrained devices, certificate chains and supply-chain dependencies. PQC is necessary, but it is far from a ‘find and replace’ as the impact and feasibility over using PQC has to be assessed.

Each of these beliefs falls apart under scrutiny, but they persist because quantum computing sits at the intersection of science, engineering, policy and imagination. People fill gaps in understanding with optimism, pessimism or mythology when it should always be evidence based.

My journey through quantum has therefore been as much about cutting through noise as it has been about learning the technology itself. I’ve had to build a grounded, evidence based understanding that resists hype, avoids fatalism and focuses on measurable reality. That mindset has shaped everything I’ve done since from how I evaluate quantum hardware, to how I design threat models, to how I communicate risk to organisations that must prepare for a future which is arriving unevenly, but undeniably.

11.4 My View of the Quantum Future

As I look ahead, my perspective on quantum computing is shaped not by hype or speculation, but by the patterns that emerge when you analyse the technology through a security and evidence driven lens. Quantum computing is transformative, but not in the way popular narratives often suggest. It will not replace classical computing. It will not become a universal machine capable of running every workload faster or more efficiently. It will not render traditional architectures obsolete.

Quantum computing will instead evolve into a specialised accelerator, much like GPUs, TPUs and other domain specific architectures that complement. Rather than replace classical systems. Its strengths lie in narrow but profoundly important domains:

Optimisation. Quantum and quantum inspired algorithms have the potential to reshape how we approach complex optimisation problems, from logistics to finance to energy systems. These are areas where even marginal improvements can have enormous real world impact.

Simulation. Quantum systems excel at simulating quantum phenomena, something classical machines struggle with fundamentally. This capability will drive breakthroughs in chemistry, physics and biological modelling, enabling discoveries that were previously out of reach.

Cryptanalysis. This is the domain with the most direct security implications. Quantum computers will not break all cryptography, but they will break the cryptographic assumptions that underpin global secure communication. This alone makes quantum a strategic capability.

Materials Science. The ability to model molecular interactions with high fidelity will accelerate the development of new materials, pharmaceuticals and energy technologies. These advances will have geopolitical, economic and security consequences.

In other words, quantum computing will not be a general purpose replacement for classical systems. It will be a powerful tool in the domains where its computational model aligns with the structure of the problem and can solve a problem a classical computer cannot.

For the security community, this distinction matters. Quantum is not a curiosity to be monitored from a distance. It is a strategic capability, one that will shape national security, economic competitiveness and the long term viability of cryptographic systems. Treating it as anything less would be a profound mistake.

The future of quantum is neither utopian nor catastrophic. It is specialised, impactful and unavoidable. Preparing for that future requires clear thinking, disciplined measurement and a willingness to confront the fragility of our current security foundations across people, process and technology.

11.5 The Security Imperative: What Must Happen Next

Understanding quantum computing is only the first step. The real challenge and the real responsibility lie in preparing for its impact. Quantum security is not something that can be deferred until hardware reaches a certain threshold or until a headline announces a breakthrough. By the time quantum machines are capable of breaking classical cryptography, it will be far too late to begin planning. Preparing for quantum requires a structured, disciplined and long term programme. Organisations

must shift from reactive security to proactive security, recognising that quantum risk is already shaping adversarial behaviour and long-term data exposure.

Several actions are essential.

- **Inventory cryptographic assets.** Most organisations do not have a clear understanding of where cryptography is used within their systems. Encryption is embedded in protocols, applications, firmware, APIs and supply chain components. Without a comprehensive inventory, migration becomes guesswork. Identifying algorithms, key lengths, certificate chains and protocol dependencies is the foundation of any quantum readiness strategy.
- **Identify long-lived data.** Not all data is equal. Some information loses value quickly; other data retains sensitivity for decades. Medical records, genomic data, financial archives, intellectual property and government communications fall into this category. These are the assets most vulnerable to HNDL attacks. Organisations must classify data based on confidentiality lifespan, not just operational importance.
- **Adopt post-quantum cryptography (PQC) standards.** The transition to PQC is no longer theoretical. NIST has have selected algorithms and vendors are beginning to integrate them. Organisations must begin adopting PQC in a phased, controlled manner. This includes hybrid approaches that combine classical and quantum resistant algorithms to ensure continuity during the transition.
- **Test PQC performance and interoperability.** PQC algorithms behave differently from classical ones, introducing larger key sizes, altered handshake patterns and distinct performance characteristics. Before deployment, organisations need to understand how these differences manifest in practice by examining latency impacts, bandwidth demands, protocol compatibility, behaviour on constrained devices and interoperability across vendors and platforms. This testing phase is essential; without it, the risk of operational disruption becomes unavoidable.

Quantum migration itself is not a patch but a multiyear transformation and organisations must approach it with structured planning rather than ad-hoc updates. Effective roadmaps prioritise high risk systems, define phased deployment strategies, coordinate with supply chain partners, update governance and compliance frameworks and ensure that technical teams receive appropriate training. A well-constructed roadmap turns quantum readiness from an abstract concern into a manageable, long-term programme of work.

At the same time, organisations must continuously monitor the progress of quantum hardware. Capability is advancing unevenly across vendors, architectures and research groups. Making it necessary to track qubit counts, error rates, coherence times, algorithmic developments and cryptanalytic milestones. This monitoring ensures that migration efforts remain aligned with real world progress rather than speculation or hype.

Finally, quantum threat modelling must be integrated into existing risk assessments. Traditional models assume classical computational limits, but quantum capability

alters those assumptions in fundamental ways. Organisations need to consider HNDL, accelerated key search capabilities, hybrid classical quantum attack chains and the supply chain vulnerabilities introduced by PQC adoption. Threat modelling must evolve alongside the technology to remain relevant and effective.

QUANTA: mapping quantum adversarial capability

While Q-SLICE was built to measure the behaviour of quantum hardware under realistic load, noise and cryptographic pressure, it quickly became clear that measurement alone was not enough. Understanding quantum systems empirically is essential, but security requires more than observation. It requires modelling the adversary their capabilities, constraints, incentives and attack pathways. This is where QUANTA emerged.

If Q-SLICE answers the question “What does quantum hardware actually do?”, then QUANTA answers the equally critical question “What can an adversary actually do with quantum capability?”

Quantum security cannot be governed by classical threat models alone. Quantum introduces new computational behaviours, new scaling patterns, and new asymmetries that reshape the economics of attack. QUANTA was designed to capture these shifts in a structured, repeatable and evidence driven manner. Helping to score and then introduce relevant controls to mitigate risk.

The purpose of QUANTA

QUANTA is a quantum-aware cybersecurity control framework and scoring system, operating in the same governance space as ISO 27001 and the NIST Cybersecurity Framework. Its purpose is to define the controls, maturity levels, and organisational capabilities required to manage the risks introduced by emerging quantum technologies.

Where classical frameworks assume stable computational limits and static cryptographic strength, QUANTA introduces quantum specific control domains, including:

- Cryptographic transition and crypto agility controls to manage the impact of quantum-accelerated cryptanalysis.
- Temporal risk and long lived data controls aligned to HNDL, HNFL, TNFL and DNEL adversary models.
- Hybrid resilience controls that account for attack chains combining classical and quantum resources.
- Resource bounded capability tracking to ensure controls scale with realistic, not hypothetical, quantum progress.
- Incremental governance, recognising that quantum capability evolves unevenly and adversaries exploit each stage.

QUANTA provides a structured, evidence driven framework for developing quantum resilient security programmes. At its core, it defines a comprehensive set of control families that span cryptography, system architecture, governance, monitoring, supply chain assurance and lifecycle management. These controls give organisations a coherent way to embed quantum-aware security practices across both technical and organisational domains.

The framework also introduces maturity tiers that assess readiness across classical, hybrid and post quantum states, allowing organisations to understand their current posture and plan their progression. Complementing this, QUANTA includes scoring mechanisms that quantify exposure, preparedness, and compliance with quantum-era requirements, enabling consistent measurement and benchmarking.

A key feature of QUANTA is its lifecycle aware approach, recognising that systems deployed today will remain operational long enough to be exposed to future quantum threats. To address this, the framework defines controls that explicitly manage long lived data risk and deferred exploitation scenarios. It also provides transition pathways that guide organisations from current state cryptography toward architectures aligned with post-quantum standards.

Quantum security is not a switch

It is a programme strategic, long term and essential. Preparing for quantum is not about predicting the exact moment when cryptographic collapse will occur. It is about ensuring that when that moment arrives, organisations are not caught unprepared. The work begins now and the organisations that act early will be the ones that maintain trust, resilience and operational continuity in the quantum era.

Quantum computing is not a distant horizon or a speculative threat. It is a present day pressure on the assumptions that underpin digital trust. My journey through quantum security has been shaped by a commitment to evidence, measurement and adversarial realism. Principles that guided the creation of the Q-SLICE threat model and QUANTA cybersecurity framework. Preparing for the quantum era is not about predicting the exact moment of cryptographic collapse; it is about building the resilience to withstand it whenever it arrives.

